AML FOOTBALL DOC’S AND USER’S MANUAL

CONTENTS

1. Mission & Purpose……..1

2. Test/Evaluation Results……..2

3. User's Manual & Guide……5

4. App Limitations & Risks……8

5. Conclusion……..9

MISSION & PURPOSE

AML Football sits at the intersection of football, machine learning, and mathematics. More than a goal prediction platform, AML Football is built on a single belief: that AI and mathematics can meaningfully change how we understand and analyse the beautiful game.

It started as a weekend project, inspired by *Moneyball*, a film about using mathematics and statistics to find hidden edges and advantages in sport. That sparked a question: could data science do the same for football? Not for gambling, but to break down what actually makes teams win, what gives them momentum, and whether the hidden patterns inside the sport we love could be unlocked with data and AI.

The problem every football fan or analyst faces is simple: we want to know what to expect from a match before it happens. Not just who might win, but the shape of the game itself, the scenarios, the possibilities, the probabilities. AML Football is built to solve that.

We are starting with goal related predictions, but the vision goes further, position predictions, player heatmap forecasting, and every meaningful way AI can be applied to football. This is version one of something much bigger.

Who It's For:

AML Football is built for football fans, analysts, data professionals, and ML/AI enthusiasts who believe the game deserves more than gut feeling or guessing and want to see what happens when AI meets football.

Test/Evaluation Results

We carried out extensive testing to evaluate how reliable the AML Football system is.

Initially, the goal was to predict match outcomes i.e win, lose, or draw. After multiple training runs and modelling approaches, every attempt fell short of a reliable accuracy threshold. Predicting match outcomes in football is notoriously difficult even for the best models in the world, and we were not going to ship something we couldn't trust.

Below are results based on test data that have not been seen by the model:

1. **Initial Accuracy for Three Way Outcome:**

   Evaluation was carried out on both Laliga and Serie A models possible 3 way outcome accuracy.

   Laliga model recorded a 42% accuracy and Serie A recorded a 47% accuracy which is slightly higher but still generally low for real world performance. This metrics tells us how accurate the model can be in real world scenario, with this current performance if a team is predicted to win it is only 42 or 47 percent sure of that prediction for Laliga and Serie a respectively.

```
5]: accuracy = accuracy_score(laliga["Actual_Outcome"], laliga["Predicted_Outcome"])
    accuracy

5]: 0.42395437262357416
```

(Laliga Accuracy Score)

```
9]: accuracy = accuracy_score(seriea["Actual_Outcome"], seriea["Predicted_Outcome"])
    accuracy

9]: 0.4742857142857143
```

(Serie A Accuracy Score)

2. Regression Metrics:

To evaluate how accurately the model predicts goals, we used two standard regression metrics:

Root Mean Squared Error (RMSE): Calculated by subtracting the predicted goals from the actual goals, squaring the result, then taking the square root to make the figures absolute. RMSE penalises large errors more heavily, meaning the model is punished significantly for predictions that are wildly off.

Mean Absolute Error (MAE): Measures the average absolute difference between predicted and actual goals. This gives a more intuitive view of how wrong the model typically is on any given prediction, think of it as the average margin of error per match.

Below shows Laliga and Serie A has a almost a similar result, it is basicly saying if the model predicts 2 goals for a team on a normal day, if the model is to be wrong the actual goal might be 3 or 1 goal.

```
Model Evaluation Metrics (per match):
-------------------------------------
Home Goals -> MAE: 0.96, RMSE: 1.20
Away Goals -> MAE: 0.99, RMSE: 1.33

Notes:
- MAE shows the typical error per match (average goals off).
- RMSE shows typical error magnitude, giving more weight to larger mistakes or model prediction error.
```

(Laliga Metrics)

```
Model Evaluation Metrics (per match):
-------------------------------------
Home Goals -> MAE: 0.86, RMSE: 1.07
Away Goals -> MAE: 0.88, RMSE: 1.11

Notes:
- MAE shows the typical error per match (average goals off).
- RMSE shows typical error magnitude, giving more weight to larger mistakes or model prediction error.
```

(Serie A Metrics)

3. Acceptable Prediction Rate:

I didn't expect us to achieve perfect accuracy because football is filled with unpredictable events that can decide how a game eventually ends. I planned to manage the errors and be able to use them to our advantage instead.

To measure how accurate the system can be, we focused on evaluating how consistent the model was in predicting actual goals, +1 goals and -1 goals.

This evaluation showed Laliga having an average of approximately 80% and Serie A having approximately 87% accuracy.

```
[61]: count = laliga["Home_Error"].isin([0, 1, -1]).sum()
      print(count/len(laliga))

      0.7965779467680608
```

(Laliga 0,+1 & -1 Error Accuracy)

```
[15]: count = seriea["Home_Error"].isin([0, 1, -1]).sum()
      print(count/len(seriea))

      0.8647619047619047
```

(Serie A 0,+1 & -1 Error Accuracy)

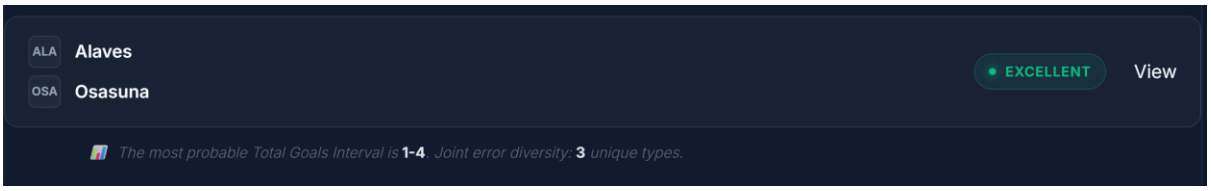
CONCLUSION:

We focused on this result as our evaluation metrics because it directly supports the statistical and mathematical way our system was built.

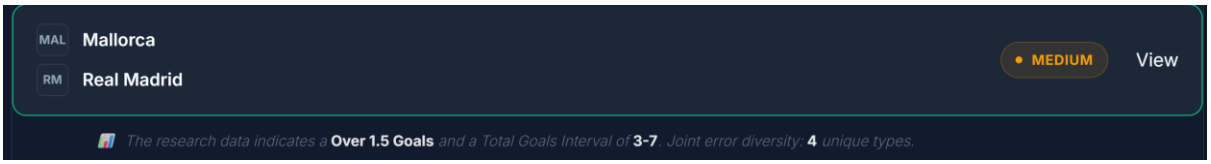
USER'S MANUAL & GUIDE

I won't go into deep technical details about how this system was built, but I will give a thorough explanation of how to analyse the data and information the app provides and a complete walkthrough of how to use it.

1. **Matches Segmentation:** There are different levels of reliability depending on the teams involved in a match, and this is what led us to segment each match based on the total combined prediction error the system has recorded for both teams.
   - **A.** Excellent Matches are matches where the model has identified a highly predictable pattern for both teams involved. These teams tend to play in a consistent, measurable way that the model reliably captures. A match is classified as Excellent when the combined count of unique prediction errors across both teams is 3 or fewer.



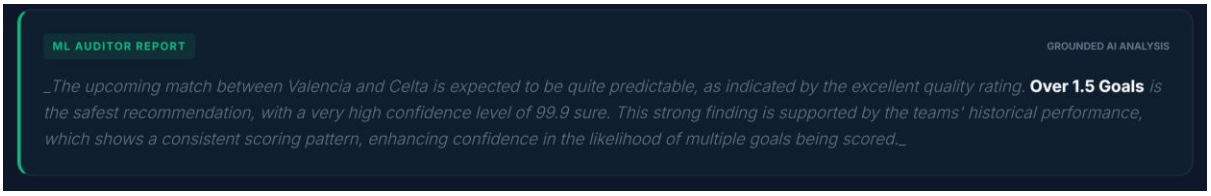
   - **B.** Medium Matches are matches that are somewhat predictable but less consistent than Excellent tier matches. They have a wider error distribution and typically carry a wider goal interval than Excellent matches, making them slightly harder to call with confidence. The combined count of unique prediction errors for this group falls between 4 and 5.



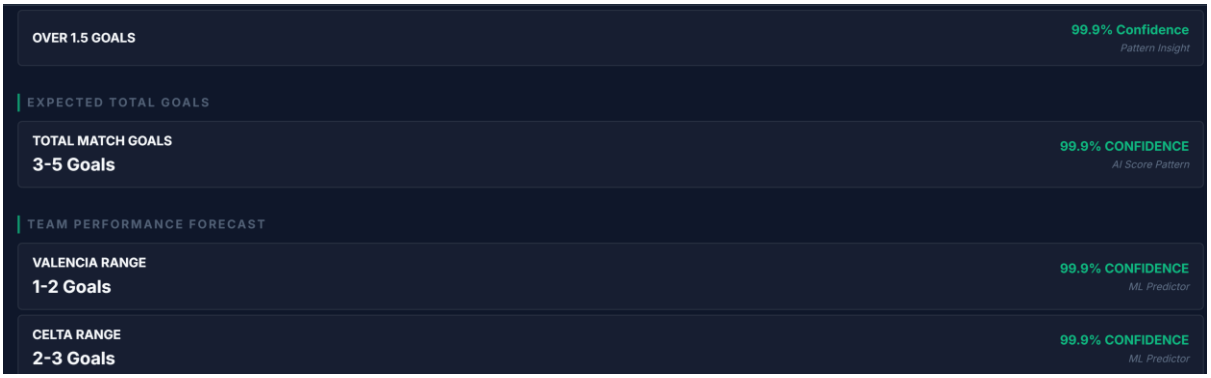
   - **C.** Low Matches consist of teams with highly unpredictable scoring patterns. Their prediction errors are more diverse and spread out than those in the Excellent and Medium tiers. Due to their unreliable nature, the system intentionally assigns them a wide goal interval such as 0–7 or 0–9 to reflect the genuine uncertainty around what can happen in these matches.

These are matches the system cannot confidently narrow down, and users should treat them accordingly. The combined count of unique prediction errors for this group is 6 or above.

2. **Match Summary:** This section contains a short most probable summary for each the entire game.
   A. AI Analyses: The AI reviews all predictions generated by the system and recommends the best and safest outcome to consider. The recommendation is driven by the percentage likelihood of the most probable scenario, giving users a single, clear insight rather than having to interpret multiple data points themselves.

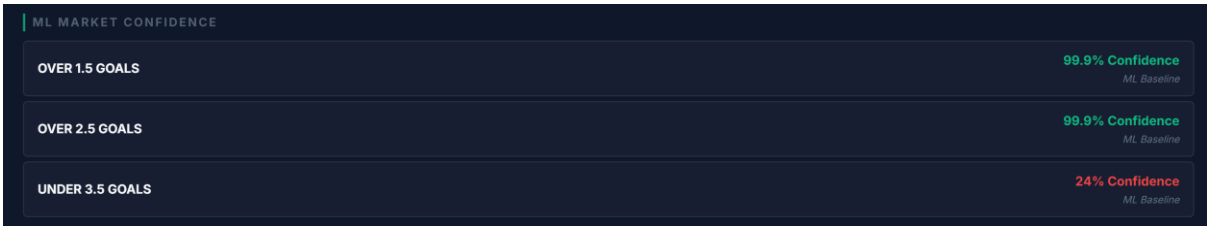


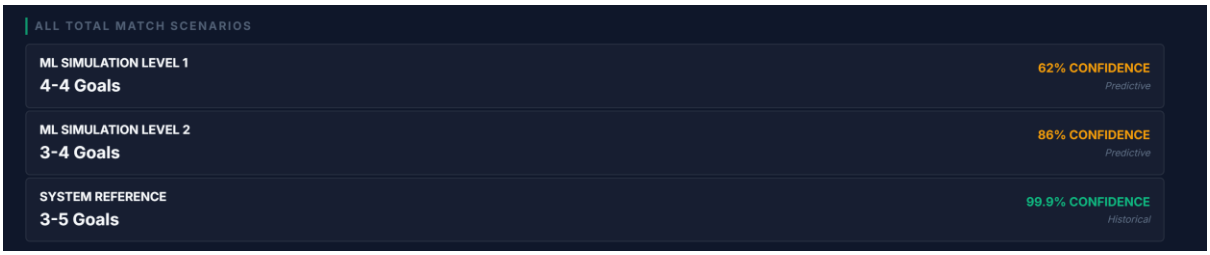
   B. Goals Prediction Summary: This section displays the most probable picks generated by the system across each prediction category. From each group of possible outcomes, the system identifies and surfaces the single most likely option covering total over/under goals, the total goals interval for the match, and the individual goal interval for each team. This gives users a clean, consolidated view of what the model considers the most realistic scoring scenarios for the match.

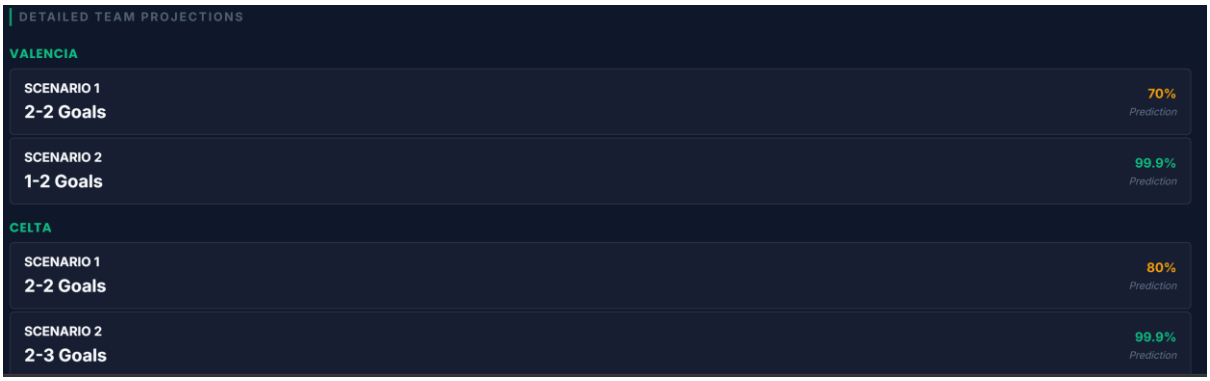


3. **Full AI Forecast**: This section contains a full breakdown of the probabilities assigned to different match scenarios. It covers over/under goals, total goal intervals, and individual team goal intervals. However, for analysis purposes,

we recommend focusing on the total goal interval rather than the individual team intervals, as it tends to be the more reliable indicator. The individual team goal intervals are better explored through the **Team Insight** section, which is dedicated to breaking down each team's goal probability separately.







ANALYSIS BREAKDOWN:

Using the images above, the probabilities can be read like this:

For The Over And Under Section: 'There is a 99.9% probability that the full match will have over 1 goal, and the same for over 2 goals. However, there is only a 24% chance that the match will not produce 4 or more goals—meaning there is a 76% chance the total goals will be 4 and above.'

For The Total Goals Interval Section: There is a 62% chance they will score exactly 4 goals, and a 86% chance they will score 3 to 4 goals but 99.9% chance they will score from 3 to 5 goals.'
Same idea for the Team Goals Interval.

APP LIMITATIONS AND RISKS

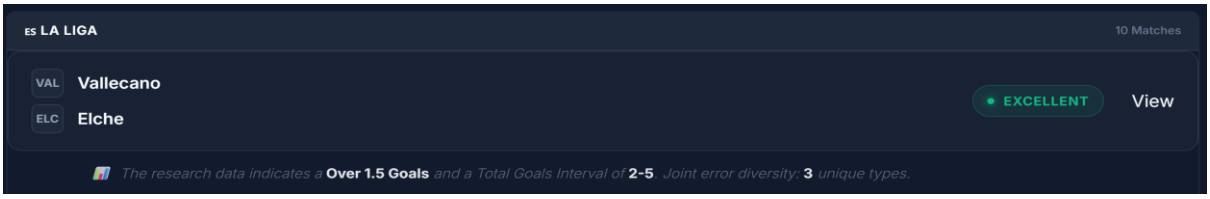
After careful analysis and monitoring, we identified that the model's reliability drops significantly when unexpected in-game events occur such as an early red card, unusual weather conditions, or an unusual goal outburst, particularly from the home team. These scenarios fall outside the patterns the model was trained to recognise and can significantly alter the natural flow of a match.

This remains one of our key challenges, and we are actively working to incorporate relevant data to better account for these situations in future iterations.

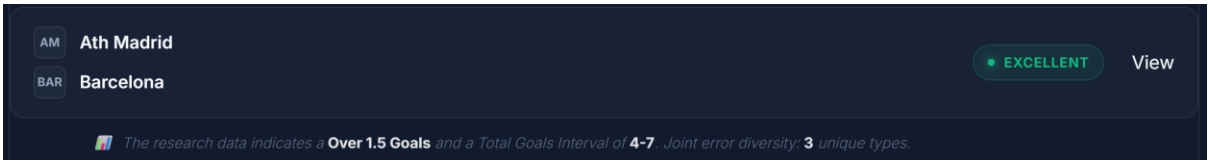
Note: Even a match classified as Excellent is not immune to these unpredictable events. The segmentation reflects historical pattern reliability, not a guarantee of outcome.

Examples Of Matches Classified As Excellent But Had Early Game Red Card:

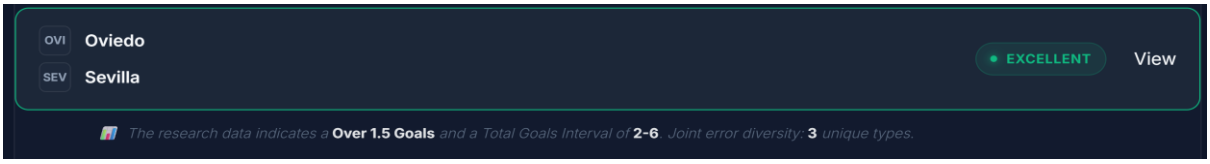
1. Vallecano vs Elche: Red card for Elche in the first half. Game ended 1-0



2. Ath Madrid vs Barcelona: Red card for Ath Madrid in the first half. Game ended 1-2



3. Oviedo vs Sevilla: Red card for sevilla in the first half. Game ended 1-0



CONCLUSION

AML Football was built on a simple but ambitious idea: that mathematics and machine learning can give football fans and analysts something they have never had before, What they should expect before a match kicks off.

Version one proves the concept. With an Acceptable Prediction Rate of approximately 80% for La Liga and 87% for Serie A. It is treated as a system built on probability, trained on data, but like any system, it has it limitations.

For this first version, performance will be judged primarily on Excellent tier matches, matches where both teams show the most consistent and predictable patterns. This gives the system the best conditions to demonstrate its true reliability before we expand evaluation across all tiers in future iterations.

This is just the beginning. Goal prediction is the foundation, but the vision for AML Football extends into position predictions, player heatmap forecasting, and every meaningful intersection between AI and the beautiful game.

The unpredictability of football is not a problem to be solved, it is what makes the sport worth watching. AML Football does not claim to remove that uncertainty. It simply helps you understand it better.